Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T SemesterV)

Applied Business

Practical – X

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Practical 10: Regression Analysis and Time Series Analysis Part A: Regression Analysis

1. Load the sales_data.csv dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
```

14.7s

```
# 1. Load dataset
df = pd.read_csv("sales_data.csv")
```

0.0s

2. Display the first five records.

```
# 2. Display first five records
print("First Five Records:")
print(df.head())

# 3 & 4. Independent and dependent variables
X = df[["advertising"]]
y = df["sales"]

print("\nIndependent Variable: Advertising")
print("Dependent Variable: Sales")
```

```
First Five Records:
   advertising  sales
0           10     15
1           12     18
2           15     20
3           18     24
4           20     27

Independent Variable: Advertising
Dependent Variable: Sales
```

3. Identify the independent and dependent variables.

```
# 3 & 4. Independent and dependent variables
X = df[["advertising"]]
y = df["sales"]
```

```
Independent Variable: Advertising
Dependent Variable: Sales
```

4. Select advertising as the independent variable and sales as the dependent variable.

```
Independent Variable: Advertising
Dependent Variable: Sales
```

5. Check the data for missing values.

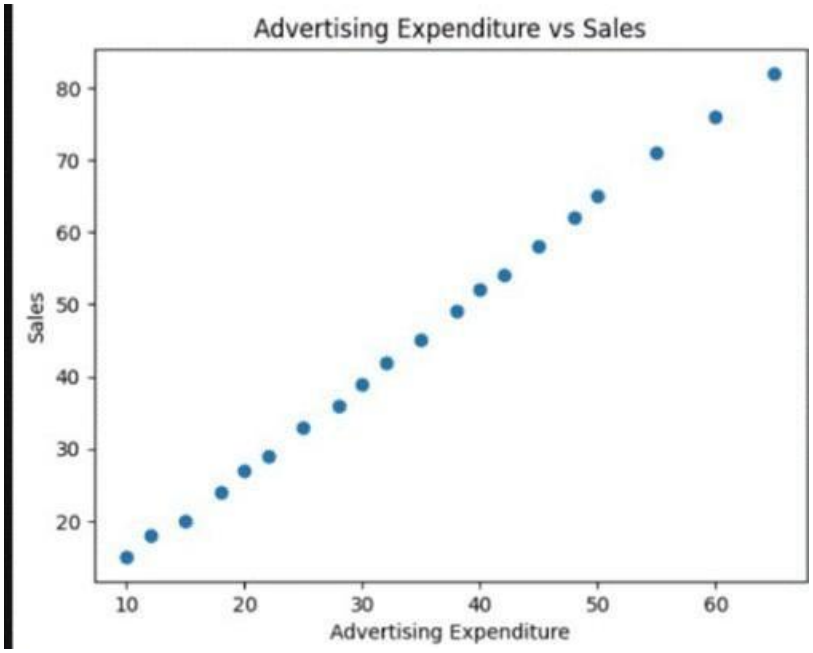
```
# 5. Check missing values
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
advertising    0
sales          0
dtype: int64
```

6. Create a scatter plot of advertising expenditure and sales.

```
# 6. Scatter plot
plt.scatter(df["advertising"], df["sales"])
plt.xlabel("Advertising Expenditure")
plt.ylabel("Sales")
plt.title("Advertising Expenditure vs Sales")
plt.show()
```

✓ 0.2s



7. Develop a simple linear regression model.

```
# 7. Develop Simple Linear Regression Model
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

# 8. Regression equation
b0 = model.intercept_
b1 = model.coef_[0]

print("\nRegression Equation:")
print(f"Sales = {b0:.2f} + {b1:.2f}(Advertising)")
```

✓ 0.4s

```
Regression Equation:
Sales = 1.75 + 1.24(Advertising)
```

8. Display the regression equation.

```
# 8. Regression equation
b0 = model.intercept_
b1 = model.coef_[0]

print("\nRegression Equation:")
print(f"Sales = {b0:.2f} + {b1:.2f}(Advertising)")
```

0.4s

```
Sales = 1.75 + 1.24(Advertising)
```

9. Use the model to predict sales for a given advertising expenditure.

```
# 9. Predict sales for a given advertising expenditure
advertising_value = 50
predicted_sales = model.predict([[advertising_value]])

print(f"\nPredicted Sales for Advertising = {advertising_value}: "
      f"{predicted_sales[0]:.2f}")

# Predictions for test data
y_pred = model.predict(X_test)
```

```
Predicted Sales for Advertising = 50: 63.96
```

10. Calculate the R^2 score of the model.

```
# 10. R2 Score
r2 = r2_score(y_test, y_pred)
print(f"\nR2 Score: {r2:.4f}")
```
✓ 0.0s

```
R2 Score: 0.9987
c:\Users\Sagar Kandari\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py:2827:
  warnings.warn(
```

11. Calculate the Mean Squared Error (MSE).

```
# 11. Mean Squared Error
mse = mean_squared_error(y_test, y_pred)
print(f"MSE: {mse:.4f}")
```

```
MSE: 0.8345
```

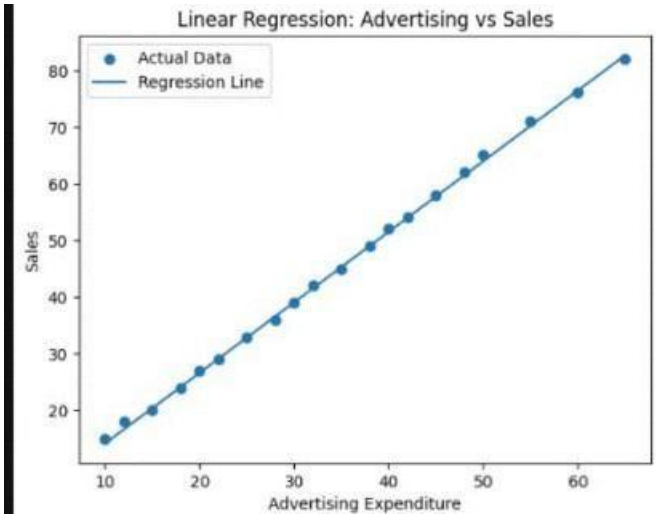
12. Compare the actual sales with the predicted sales

```
Actual vs Predicted Sales:
   Actual Sales  Predicted Sales
0            15        14.191227
1            71        70.184601
2            62        61.474521
3            18        16.679822
```

.

13. Plot the regression line with the actual data points.

```
# 13. Plot regression line with actual data
plt.scatter(X, y, label="Actual Data")
plt.plot(X, model.predict(X), label="Regression Line")
plt.xlabel("Advertising Expenditure")
plt.ylabel("Sales")
plt.title("Linear Regression: Advertising vs Sales")
plt.legend()
plt.show()
```



14. Interpret the regression results.

```
# 14. Interpret regression results
print("\nInterpretation:")
print(f"For every 1 unit increase in advertising expenditure, "
      f"sales increase by approximately {b1:.2f} units.")

print(f"The R2 score is {r2:.4f}, which shows how well advertising "
      f"explains the variation in sales.")
```

0.1s

```
Interpretation:
For every 1 unit increase in advertising expenditure, sales increase by approximately 1.24 units.
The R2 score is 0.9987, which shows how well advertising explains the variation in sales.
```

15. Write a conclusion about how advertising expenditure is related to sales. A simple regression model is:

```
# 15. Conclusion
print("\nConclusion:")
print("Advertising expenditure has a positive relationship with sales.")
print("As advertising expenditure increases, sales generally increase.")
```

0.0s

```
Conclusion:
Advertising expenditure has a positive relationship with sales.
As advertising expenditure increases, sales generally increase.
```

$= 0 +$

$1_{()}$

:

16. Load monthly sales data.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt


✓ 7.7s
```

17. Convert the date column into date format.

```
# Step 2: Create the monthly sales dataset
data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
              "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"],
    "Sales": [120, 135, 128, 145, 150, 160,
              155, 170, 165, 180, 190, 200]
}

df = pd.DataFrame(data)

print("Monthly Sales Data:")
print(df)
```

```
Monthly Sales Data:
    Month  Sales
0     Jan    120
1     Feb    135
2     Mar    128
3     Apr    145
4     May    150
5     Jun    160
6     Jul    155
7     Aug    170
8     Sep    165
9     Oct    180
10    Nov    190
11    Dec    200
```

18. Plot monthly sales using a line chart.

```
# Step 3: Convert Month column into date format
df["Date"] = pd.to_datetime(df["Month"] + "-2025",
                            format="%b-%Y")

df.set_index("Date", inplace=True)

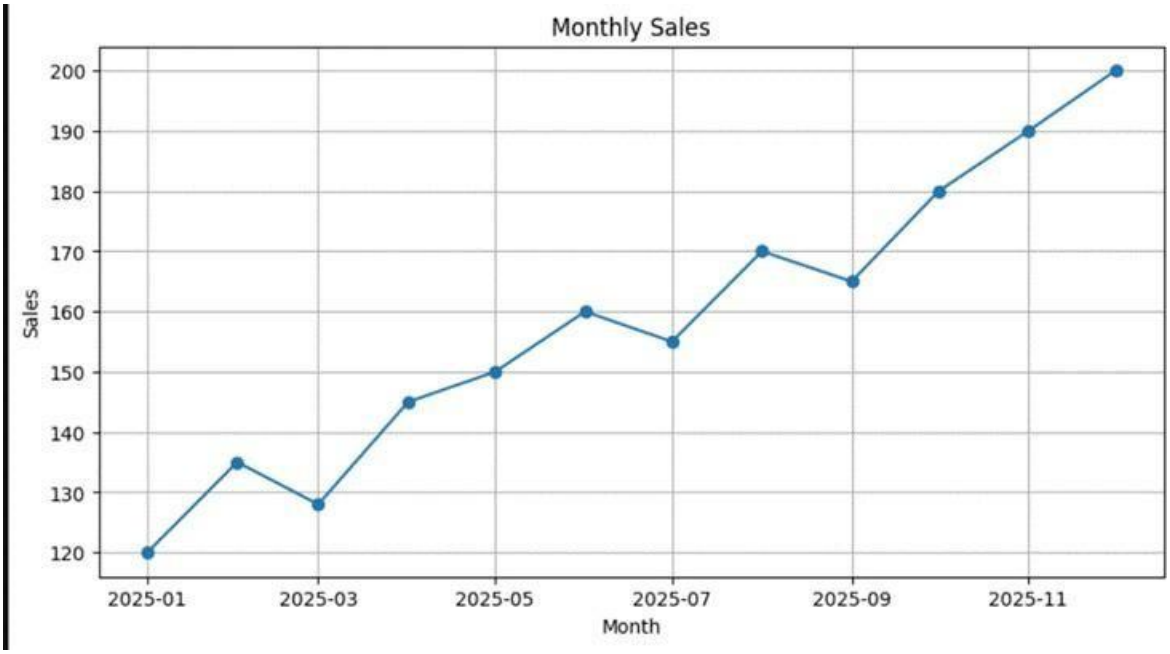
print("\nData after converting to date:")
print(df)
```

0.0s

```
Data after converting to date:
            Month  Sales
Date
2025-01-01   Jan    120
2025-02-01   Feb    135
2025-03-01   Mar    128
2025-04-01   Apr    145
2025-05-01   May    150
2025-06-01   Jun    160
2025-07-01   Jul    155
2025-08-01   Aug    170
2025-09-01   Sep    165
2025-10-01   Oct    180
2025-11-01   Nov    190
2025-12-01   Dec    200
```

19. Observe the trend.

```
# Step 4: Plot monthly sales using line chart
plt.figure(figsize=(10, 5))
plt.plot(df.index, df["Sales"], marker="o")
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



20. Apply Exponential Smoothing.

```
# 5. Apply Exponential Smoothing
es_model = ExponentialSmoothing(
    df["Visitors"],
    trend="add",
    seasonal=None
)

es_fit = es_model.fit()


# 6. Forecast next 3 months
es_forecast = es_fit.forecast(3)
```

```
Exponential Smoothing Forecast:
2026-01-01    1603.333352
2026-02-01    1654.487198
2026-03-01    1705.641045
Freq: MS, dtype: float64
```

21. Forecast the next 3 months.

```
# Step 7: Create future dates
future_dates = pd.date_range(
    start=df.index[-1] + pd.DateOffset(months=1),
    periods=3,
    freq="MS"
)
```

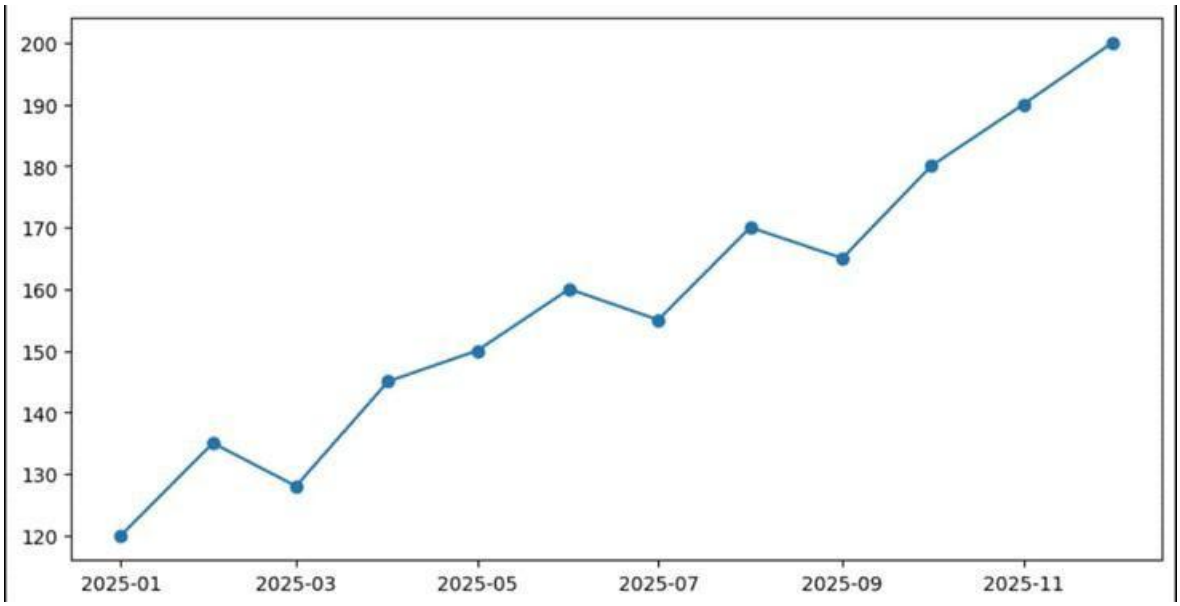
22. Plot actual and forecasted sales.

```
plt.figure(figsize=(10, 5))

plt.plot(df.index, df["Sales"], marker="o", label="Actual")
plt.plot(future_dates, forecast_es, marker="o", label="Exponential Smoothing")
plt.plot(future_dates, forecast_arima, marker="o", label="ARIMA")

plt.title("Sales Forecast")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.legend()
plt.show()
```



23. Calculate forecasting error.

```
# 12. Exponential Smoothing on training data
es_test_model = ExponentialSmoothing(
    train,
    trend="add",
    seasonal=None
)

es_test_fit = es_test_model.fit()

es_test_forecast = es_test_fit.forecast(3)
```

```
Exponential Smoothing:
MAE = 74.67
RMSE = 78.96
```

24. Apply ARIMA to the same data.

```
# 18. Display next 3 months forecast
forecast_table = pd.DataFrame({

    "Month": future_dates.strftime("%b"),

    "Exponential Smoothing":
        np.round(es_forecast.values, 2),

    "ARIMA":
        np.round(arima_forecast.values, 2)
})
```

```
st -----
hing    ARIMA
3.33  1608.71
4.49  1609.66
5.64  1609.76
```

25. Identify which model gives the lower forecasting error for the given dataset. 29. Write the final conclusion.

```
# 19. Final conclusion
if es_mae < arima_mae:

    print("\nFinal Conclusion:")
    print("Exponential Smoothing gives lower forecasting error.")
    print("Therefore, it performs better for this website visitor dataset.")

else:

    print("\nFinal Conclusion:")
    print("ARIMA gives lower forecasting error.")
    print("Therefore, it performs better for this website visitor dataset.")
```

```
ives lower forecasting error.
better for this website visitor dataset.
13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so infer
s, freq)
13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so infer
s, freq)
13/dist-packages/statsmodels/tsa/base/tsa_model.py:480: ValueWarning: No frequency information was provided, so infer
s, freq)
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s, freq)
```